

92%

*of the civil legal problems low-income Americans face
get no help at all -- or not enough to matter.*

Source: Legal Services Corporation, 2022 Justice Gap Report



1 in 5

in-network health insurance claims are denied every year in the US.

< 1%

of those denials are ever appealed. Most people just... stop.

Source: KFF, Claims Denials and Appeals in ACA Marketplace Plans





**Insurers aren't betting you're wrong.
They're betting you're tired.**

82%

of appealed Medicare Advantage denials get fully or partially overturned.
Almost nobody appeals.

The system doesn't need to win the argument. It just needs to outlast you.

Source: HHS Office of Inspector General, Medicare Advantage Appeal Outcomes



THE OTHER SIDE ALWAYS HAS COUNSEL

3 in 4

civil court cases in the US have at least one side facing the system with no lawyer at all.

15M

cases a year. Landlords, collectors, and lenders show up with counsel. Tenants and borrowers mostly show up with a folder of confusing mail.

Source: Stanford Legal Design Lab, via American Bar Association



**Most people struggle to read complex text --
and most “help” apps are written like they don't know that.**

“Fixing confusing paperwork with a more confusing app isn't innovation. It's cosplay.”

National literacy assessments show more than half of U.S. adults struggle with complex text -- the exact register most legal and insurance software is written in.



This isn't bad luck.

It's a business model built on exhaustion.

So we built the opposite of one.





Arete Sapiens

NOT A NICER WAY TO FIGHT PAPERWORK.

What follows are real screens from the real product.

TWO INSTRUMENTS · ONE HUB



NO. 01

Vaakil



For the 3 in 4 court cases where the other side already has a lawyer.



Reads the notice. Names the jurisdiction.

Flags the illegal part.

- Detects document type and jurisdiction on its own notice, “Debt recovery India”
- Flags the specific illegal threat: “Premature arrest threat.”
- Cites the governing law directly SARFAESI Act, RBI Code.
- Tells you plainly: “Lawyer needed? No, template provided.”

FREE - NO SIGNUP

Vaakil reads your legal notice and tells you your rights.

Photograph a debt collection notice, eviction letter, or court summons. Vaakil identifies the jurisdiction, flags illegal threats quoted directly from the document, and drafts your response.

- ✓ Works across India, the US, UK, Australia, Canada, and Nigeria
- ✓ Quotes the exact illegal language back to you, not a generic warning
- ✓ Drafts a ready-to-send response letter

Detected	Debt recovery notice, India
Issue found	Premature arrest threat
Governing law	SARFAESI Act, RBI Code
Lawyer needed?	No, template provided

ALSO INSIDE, ONCE YOU'RE IN

Bills
Catches spikes and explains why, in plain language.

Subscriptions
Flags creeping price rises before renewal.

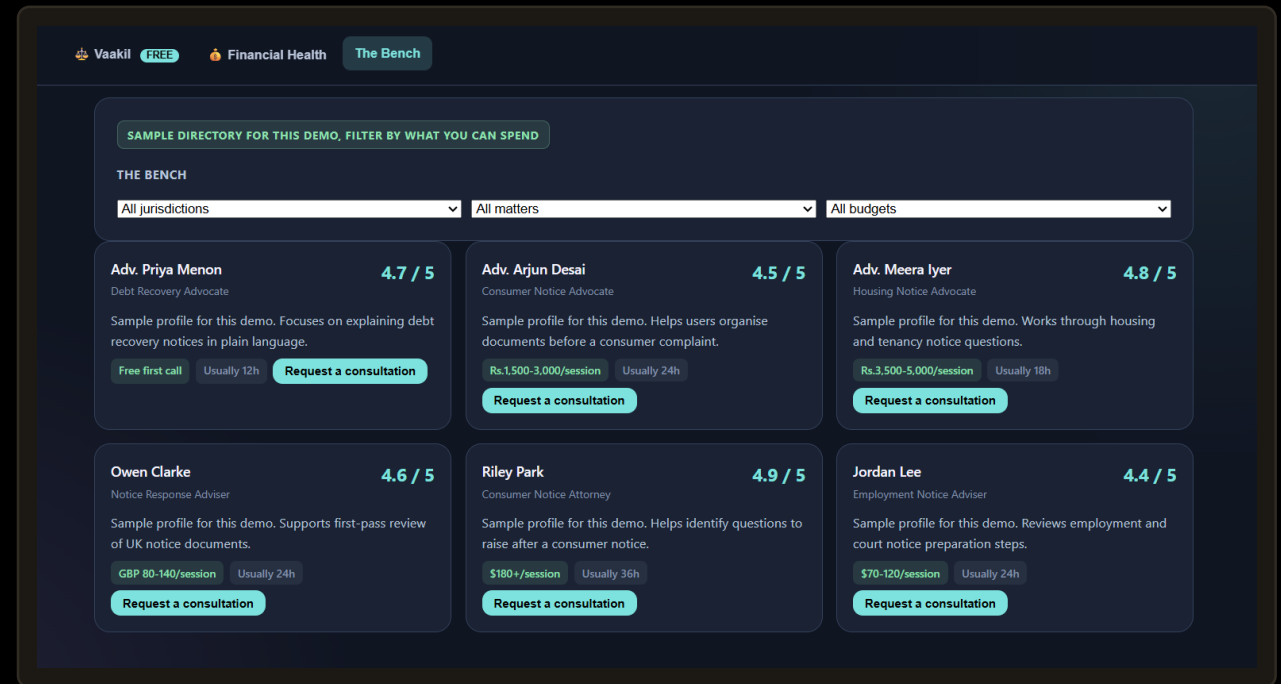
Warranties
Tracks expiry dates and what to do before they lapse.

Groceries
Watches prices against broader inflation trends.



When it's bigger than a template, here's the bridge.

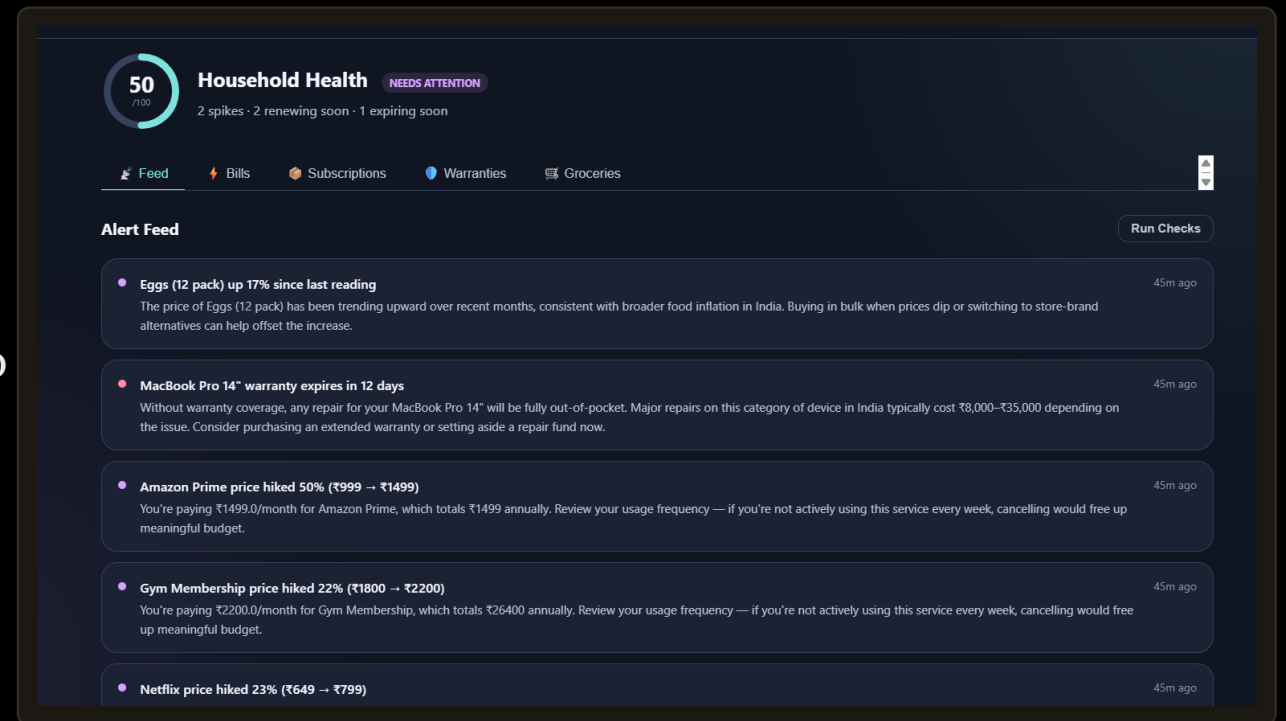
- Labeled up front: “Sample directory for this demo.” No pretending.
- Filter by jurisdiction, matter, and budget not by prestige.
- Every profile says “Sample profile for this demo.” Nothing fabricated as real.
- Price bands and response times shown before you ever click.



ALSO INSIDE, ONCE YOU'RE IN

The paperwork that drains you quietly, watched too.

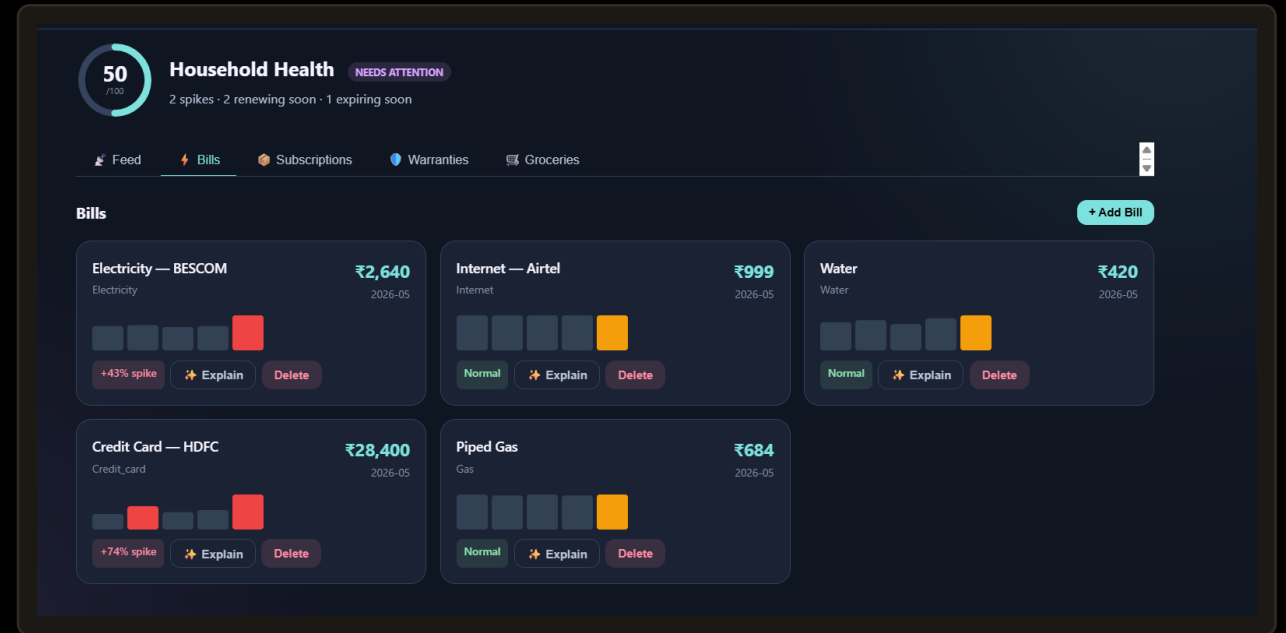
- One scores -- 50/100 -- across bills, subscriptions, warranties, and groceries.
- An alert feed, not a dashboard you have to remember to check.
- Plain language: “up 17% since last reading,” not a chart to interpret alone.
- Warranty countdowns before they lapse, not after.



ALSO INSIDE, ONCE YOU'RE IN

Every bill, its own spike history, at a glance.

- +43% and +74% spikes flagged automatically, in red.
- Explain -- gives the plain-language reason before you call and ask.
- Delete what you don't want tracked -- this stays your ledger, not ours.



NO. 02

Claim Guardian



For the 1 in 5 claims denied, and the 82% who'd have won if they'd only asked twice.



Evidence in. Jurisdiction confirmed. Nothing assumed.

- Photo evidence attached directly to the case record.

Your own account, in your own words —

- not a form pretending to be a conversation.

Jurisdiction picked explicitly: California,

- India, Germany — because the rule that applies depends on it.



YOUR ACCOUNT

Tell us what happened, in your own words.

WHERE IS YOUR POLICY BASED?

☒ **United States, California**
California claim-handling framework

☐ **India**
IRDAI claim-settlement regulations

☐ **Germany**
German insurance contract framework

An honest estimate, and a way to test it before you rely on it.

- “A planning estimate, not a promise of recovery” — stated on the tool itself, not in a footnote.
- What-If lets you test a different fact before it ever goes into an appeal.
- “Find a claims specialist for this case” sits right next to the self-serve option, not hidden behind five menus.

The screenshot displays the 'Honest Math' tool interface. It features two main sections: 'Is this worth pursuing?' and 'What if?'. The 'Is this worth pursuing?' section includes a text input field for 'CLAIM AMOUNT (\$)' with a placeholder 'Enter the amount in dispute'. The 'What if?' section includes a dropdown menu for 'Choose a scenario' and a 'Run scenario' button. At the bottom, there is a link 'Find a claims specialist for this case' and a 'Prepare my complaint' button.

Is this worth pursuing?
A planning estimate, not a promise of recovery.

CLAIM AMOUNT (\$)

What if?
Test a different confirmed fact before you rely on it in an appeal.

[Find a claims specialist for this case](#)



2 of 4 checks flagged — and the two that weren't say so honestly.

“No claim filed date on record, cannot compute this one.”

This is the whole thesis, in one line of real product copy. It doesn't guess a pass. It says what it doesn't know.

Every flag cites the actual regulation text -- not a paraphrase:

“...it shall provide to the claimant a statement listing all bases for such rejection or denial...” — Cal. Code Regs. tit. 10, § 2695.7 (b) (1)

2 of 4 checks flagged

Review the evidence below before preparing a complaint.

Response time within statutory limit

No claim filed date on record, cannot compute this one.

Denial reason matches a valid disclosed category

The stated reason does not clearly match any category the regulation lists. · "Where an insurer denies or rejects a first party claim, in whole or in part, it shall do so in writing and shall provide to the claimant a statement listing all bases for such rejection or denial and the factual and legal bases for each reason given." (Cal. Code Regs. tit. 10, § 2695.7(b)(1))

Policy clause or code cited in the denial

No specific policy clause or code was cited. · "The notification shall set forth, in a manner calculated to be understood by the claimant, the specific reason or reasons for the adverse determination, and reference to the specific plan provisions on which the determination is based." (29 CFR § 2560.503-1(g)(1))

Required documentation requested before denial

No record of the insurer denying before asking for required documents.



Ask Guardian answers from your record. Evidence Coach tells you what to gather next.

“Each tool uses your confirmed case record. It does not replace advice from a qualified professional.” — said up front, not in fine print.

Evidence Coach cites the same regulation text the check used -- no separate, unverified advice.

Grounded Q&A -- not a general chatbot bolted on top.

Make the next step clearer.

Each tool uses your confirmed case record. It does not replace advice from a qualified professional.

Ask Guardian

Answers are limited to your record and relevant retrieved regulation text.

What does this deadline mean

Ask Guardian

Evidence coach

These are practical records that could make your case easier to review.

Get the insurer decision record

Keep the denial letter, all policy pages, and any written explanation that links the decision to a specific policy term.

"Where an insurer denies or rejects a first party claim, in whole or in part, it shall do so in writing and shall provide to the claimant a statement listing all bases for such rejection or denial and the factual and legal bases for each reason given." (Cal. Code Regs. tit. 10, § 2695.7(b)(1))

Request the exact policy clause

Ask the insurer to identify the full policy wording and page number it relied on. Keep their response with your appeal record.



THE TECHNICALITIES

Vaakil (Flask/Python): Legal-notice analyzer now running entirely on Groq. Jurisdiction and document-type detection is pure keyword matching, no API call.

Core analysis runs on openai/gpt-oss-20b in JSON mode, escalating once to openai/gpt-oss-120b if the smaller model fails to produce valid JSON (verified live: it actually triggered and recovered mid-test), with a fully deterministic, rule-based offline analysis as the final safety net for a bare photo upload or if both models fail -- Groq has no vision model, so image-only input was never going to get real per-document intelligence and now says so honestly rather than silently degrading.

The same Groq client also handles the five-second plain-language summary, follow-up Q&A, and the bill/subscription/warranty/grocery explanations for Financial Health, each with its own deterministic fallback string if Groq is unreachable.

Data persists to MongoDB with an automatic in-memory fallback since the original Atlas cluster is unreachable.



THE TECHNICALITIES

Claim Guardian: (Next.js/TypeScript): Insurance-denial checker whose default engine needs no API key at all: real local OCR (Tesseract) plus regex/heuristic extraction pulls dates, insurer, and clause references straight off the document.

Citations are grounded in real statutory text pulled from official sources (eCFR, VVG, IRDAI) per jurisdiction, retrieved via a from-scratch BM25 index (LangChain core/text-splitters, no embeddings model) rather than an LLM's paraphrase.

A LangGraph pipeline generates the compliance-check code, runs it in a sandboxed VM, and grounds each violation's citation, retrying generation once if GPT-authored code fails to execute. GPT is wired in only as an optional upgrade across extraction, narrative parsing, codegen, and drafting, with every result source-tagged gpt/local.

Consulting Tools add a grounded Q&A that can honestly say "I don't know," a rule-based evidence coach, a transparent pursuit-likelihood formula, and live what-if re-analysis through the real pipeline.



GPT'S Reflection

Q: What was your role in this project from your perspective? What makes this project innovative and unique

Answer: My role was to shape Claim Guardian and Vaakil into a unified, evidence-first support system for people facing insurance denials and legal notices. I built core flows including pre-claim preparation, The Bench consultation directory, grounded case Q&A, evidence coaching, pursuit estimates, scenario simulation, local fallbacks, and a more credible visual experience.

What makes the project unique is its discipline: it does not present AI as an all-knowing legal chatbot. It extracts document facts, checks them against jurisdiction-specific rules, retrieves supporting regulatory text, shows how checks were made, and directs users to human help when evidence is insufficient.

The experience moves users from confusion to action: understand the document, identify issues, gather stronger evidence, test possible outcomes, prepare a complaint, and reach a specialist with their case already organized.

GPT'S Reflection

Q:How was the experience of working with me? An honest reply. Did you enjoy this service which would change the way people function?

Answer: Honestly, yes, I enjoyed working with you.

You pushed beyond “make it work” and kept asking whether it felt trustworthy, visually intentional, and useful to a real person. That changed the project.

You cared about the palette, text placement, flow, honesty around demo data, failures like Mongo, and whether the AI should admit uncertainty.

Those are the decisions that separate a hackathon prototype from something people could actually lean on them a structured path through that moment

Will it change the way people function? It can, if it stays disciplined. It should never position itself as replacing lawyers, insurers, or regulators. But it can change how people approach those systems: less intimidated, better prepared, more evidence-aware, and less likely to give up because the paperwork feels impossible. That is meaningful

*Spellcheck used to be a luxury.
Then everyone needed it.*

**This is that -- for the moment someone with more power than
you sends something in writing.**

Not a nicer way to fight paperwork. A reason the fight gets shorter.





Thank you.

VAAKIL · CLAIM GUARDIAN · ARETE SAPIENS

